# Open-Source Implementation of X-Nuclear Sequences Using the Pulseq Framework

**Xiaoxi Liu, Di Cui**

**02-26-2026**

# Background and Motivation

## Pulse sequence programming : low-level programming / vendor-specific

- Problem of translation across different vendors, sites, and even software versions

## Pulse sequence tools

- Interpreted to different vendors

Pulseq

VISTA RTHawk

TOPPE

pypulseq

gammaSTAR

## Pulseq in X-Nuclear MRI

- Siemens Scanner – lower-level sequence programming (e.g. X-EPI [1])
- GE Scanner – system-level sequence programming (TOPPE - only support $^{1}$H MRI)

[1] Blazey, T., Shaw, A. and von Morze, C., 2024. *MRM*, *92*(2), pp.772-781.

# Background and Motivation

Limited available subjects and data sets - a key challenge in X-nuclear MRI

- Standard acquisition for multiple scanners – multisite study

Common sequences in X-nuclear/HP MRI

- Chemical shift imaging / MR spectroscopy imaging
- Metabolite-specific imaging

**Aim: To develop a GE interpreter with X-nucleus capabilities (TOPPE MNS) and design modular sequences supporting several sequence formats.**

# C13 sequences — Modular C13 sequence in Pulseq

C13_sequences_Pulseq Public

Pin | Unwatch 1

main | 1 Branch | 0 Tags | Go to file t | Add file | Code

dcui10 Update write_C13_MS_bSSFP.m f113fc9 · 9 months ago 16 Commits

| | | |
|---|---|---|
| C13_2D_GRE_spiral | Update write_2DGRE_C13.m | 9 months ago |
| C13_MS_3DbSSFP | Update write_C13_MS_bSSFP.m | 9 months ago |
| GE interpreter with MNS function | TOPPE interpreter | 2 years ago |
| LICENSE | Update LICENSE | 2 years ago |
| README.md | Update README.md | 9 months ago |

README License

## C13_sequences_Pulseq

Modular C13 sequences designed with Pulseq framework

If you find them useful in your research, please cite:

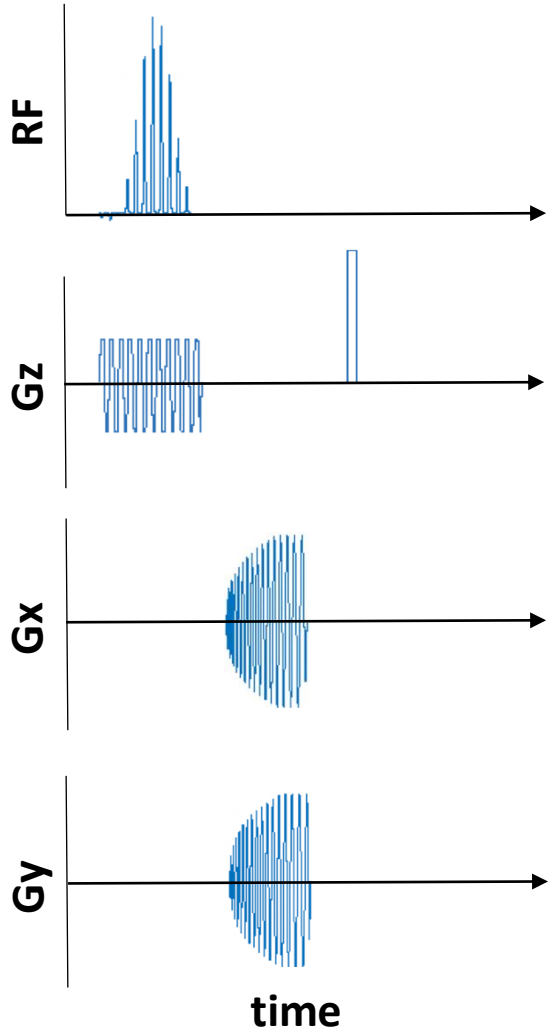
Liu, Xiaoxi, Di Cui, Peder EZ Larson, Dirk Mayer, Andreas Korzowski, Jon-Fredrik Nielsen, Rolf F. Schulte et al. "Open-source implementation of X-nuclear sequences using the Pulseq framework." Magnetic resonance in medicine (2025).

```
%% Set system limits (update these values with configuration of the scanner)
sys = mr.opts('maxGrad', 50/sqrt(2), 'gradUnit','mT/m', ...
'maxSlew', 200/sqrt(2), 'slewUnit', 'T/m/s', ...
'rfDeadTime', 100e-6, ...
'rfRingdownTime', 60e-6, ...
'rfRasterTime', 4e-6, ...
'adcDeadTime', 20e-6, ...
'adcRasterTime', 2e-6, ...
'gradRasterTime', 4e-6, ...
'blockDurationRaster', 4e-6, ...
'gamma',10708000, ...
'B0', 3.0);
```

Updating gamma for other nuclei

# C13 sequences — Modular 2D GRE spiral sequence

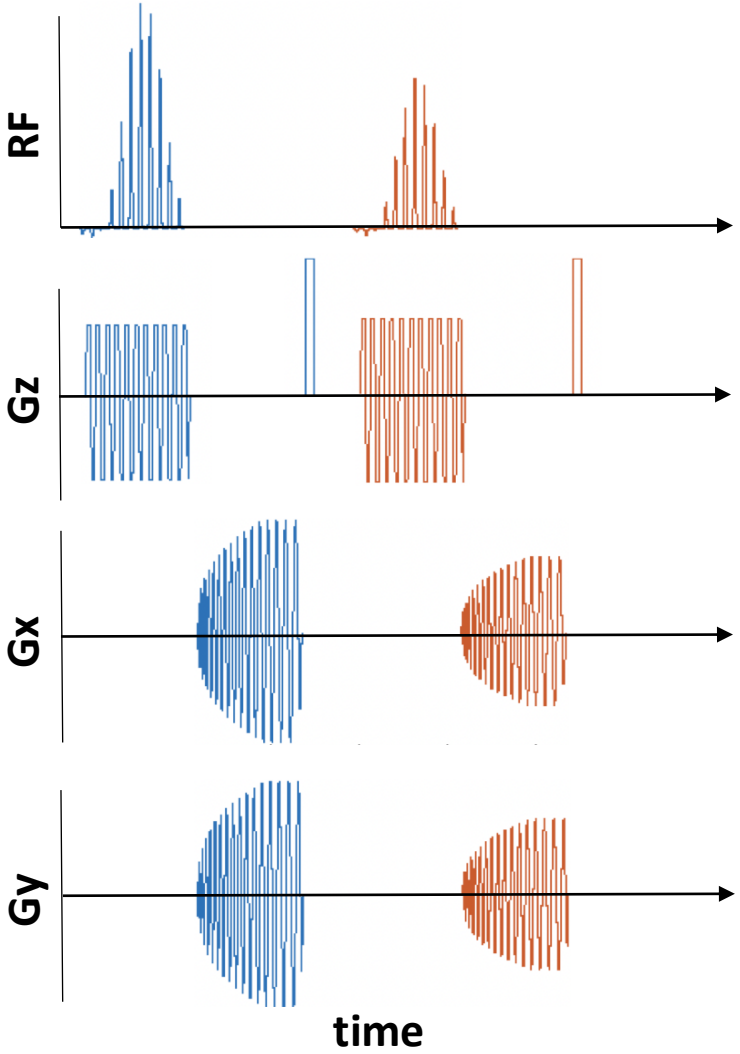
## Modular GRE spiral




## Set sequence parameters

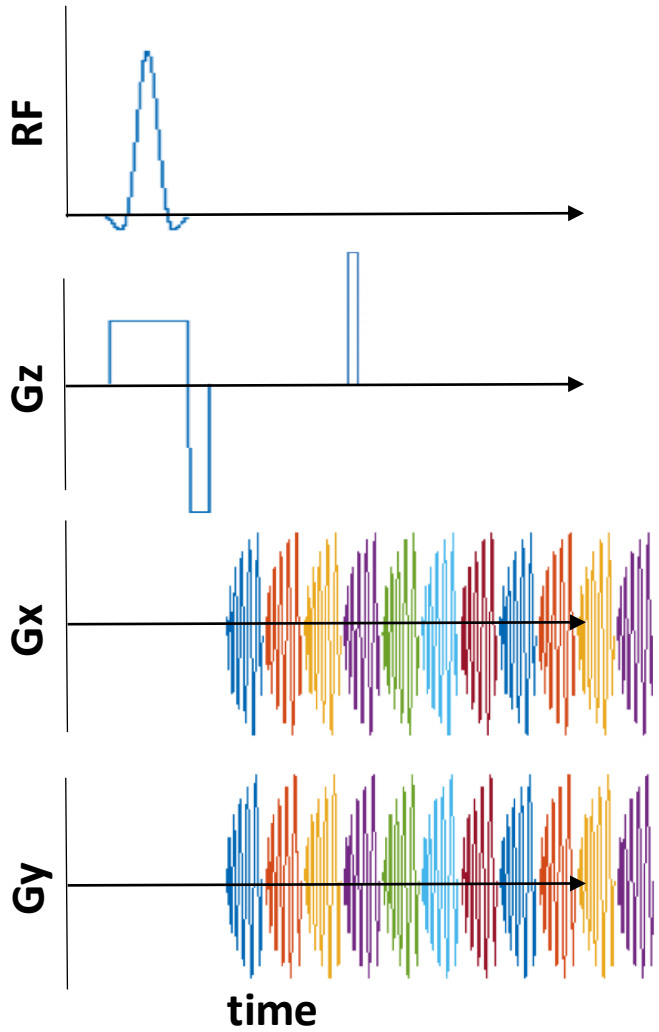
```
para = [];
para.nmet = 2; % # of metabolites
para.freqshift = [0, 390]; % Hz, frequency shift
para.res = [5, 7.5]*1e-3; % m, resolution
para.nx = 40;  % matrix size
para.sth = 15e-3;  % m, slice thickness
para.nz = 1;  % number of slice
para.necho = [1, 1];  % echo number
para.nT = 10; % number of time points
para.TR = [60, 60]*1e-3; % s
para.FA = [60, 45]; % degree, flip angle
```

## Metabolite-specific GRE (e.g.: 2 mets)



# C13 sequences — Modular 2D GRE spiral sequence

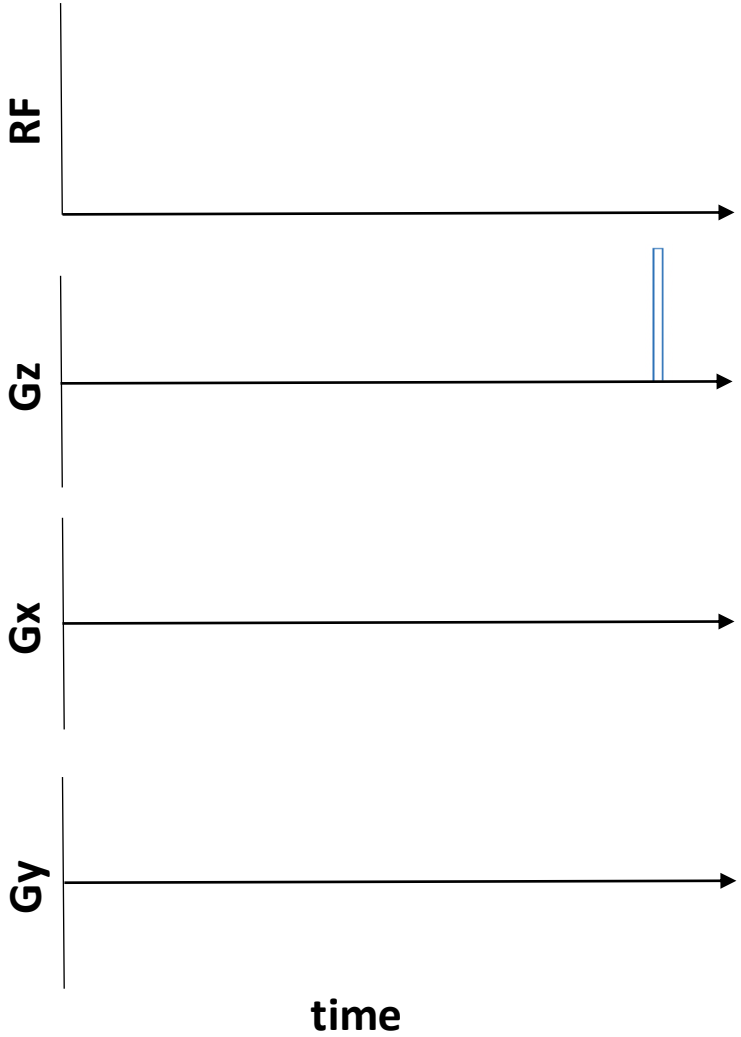
## Modular GRE spiral




## Set sequence parameters

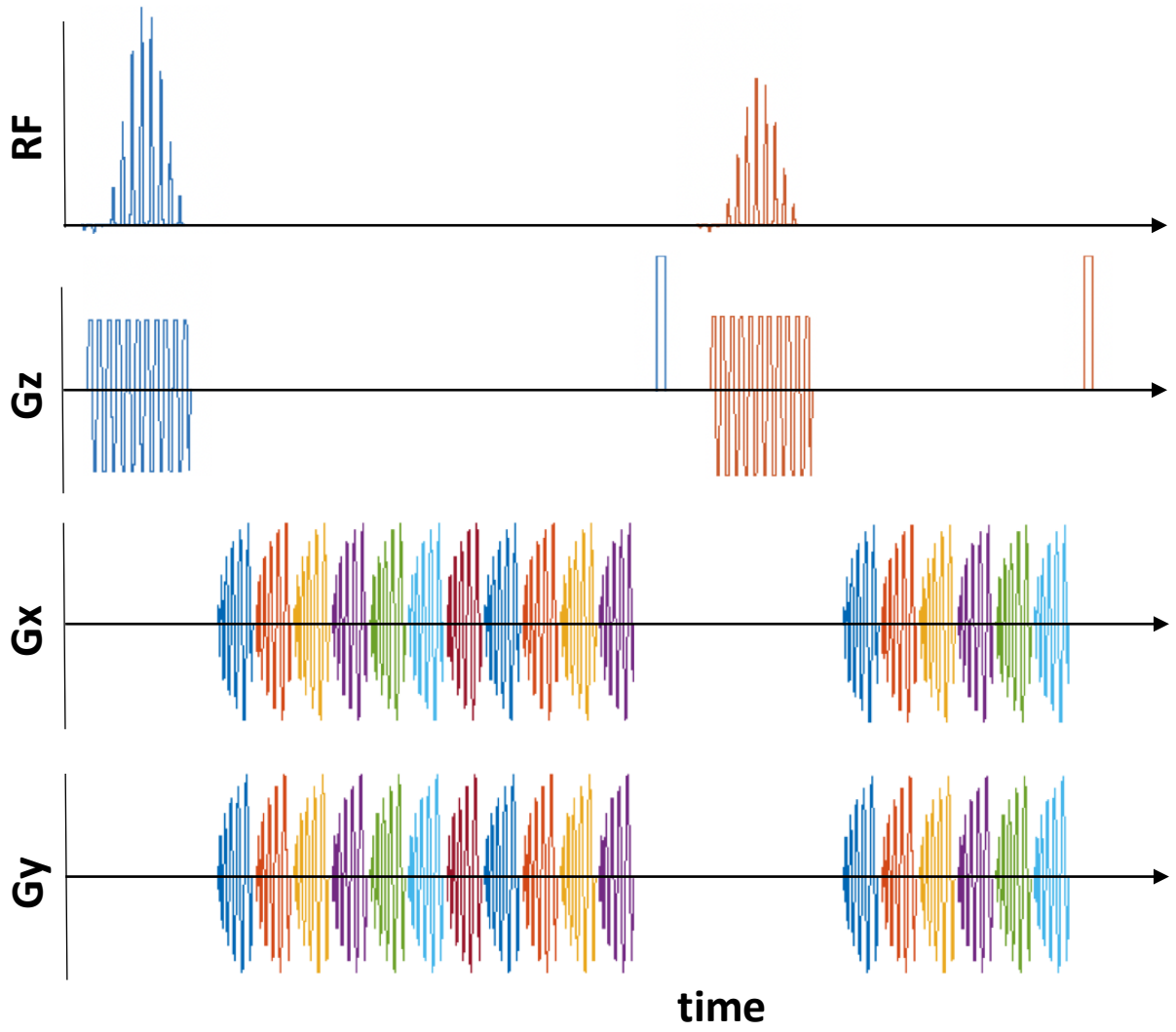
```
para = [];
para.nmet = 1; % # of metabolites
para.freqshift = [0]; % Hz, frequency shift
para.res = [5]*1e-3; % m, resolution
para.nx = 40;  % matrix size
para.sth = 15e-3;  % m, slice thickness
para.nz = 1;  % number of slice
para.necho = [10];  % echo number
para.deltaTE = 22.9e-3; % s, echo spacing
para.nT = 10; % number of time points
para.TR = [500]*1e-3; % s
para.FA = [60]; % degree, flip angle
load('sinc_pulse.mat');
```

## Chemical Shift Imaging / MRSI



# C13 sequences — Modular 2D GRE spiral sequence

Metabolite-specific multi-echo GRE (e.g.: 2 met)



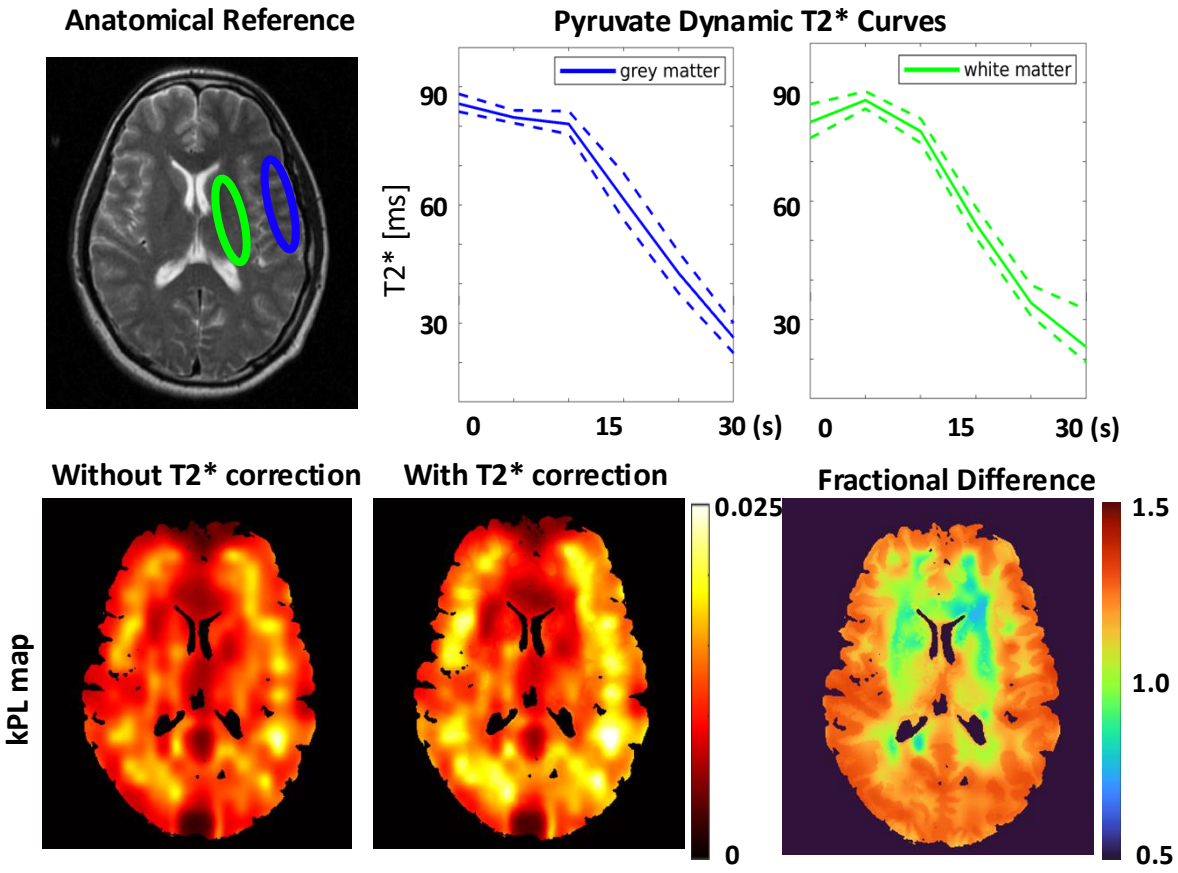

## Set sequence parameters

```
para = [];
para.nmet = 2; % # of metabolites
para.freqshift = [0, 390]; % Hz, frequency shift
para.res = [5, 7.5]*1e-3; % m, resolution
para.nx = 40;  % matrix size
para.sth = 15e-3;  % m, slice thickness
para.nz = 1;  % number of slice
para.necho = [10, 5];  % echo number
para.deltaTE = 22.9e-3; % s, echo spacing
para.nT = 10; % number of time points
para.TR = [500, 250]*1e-3; % s
para.FA = [60, 45]; % degree, flip angle
```

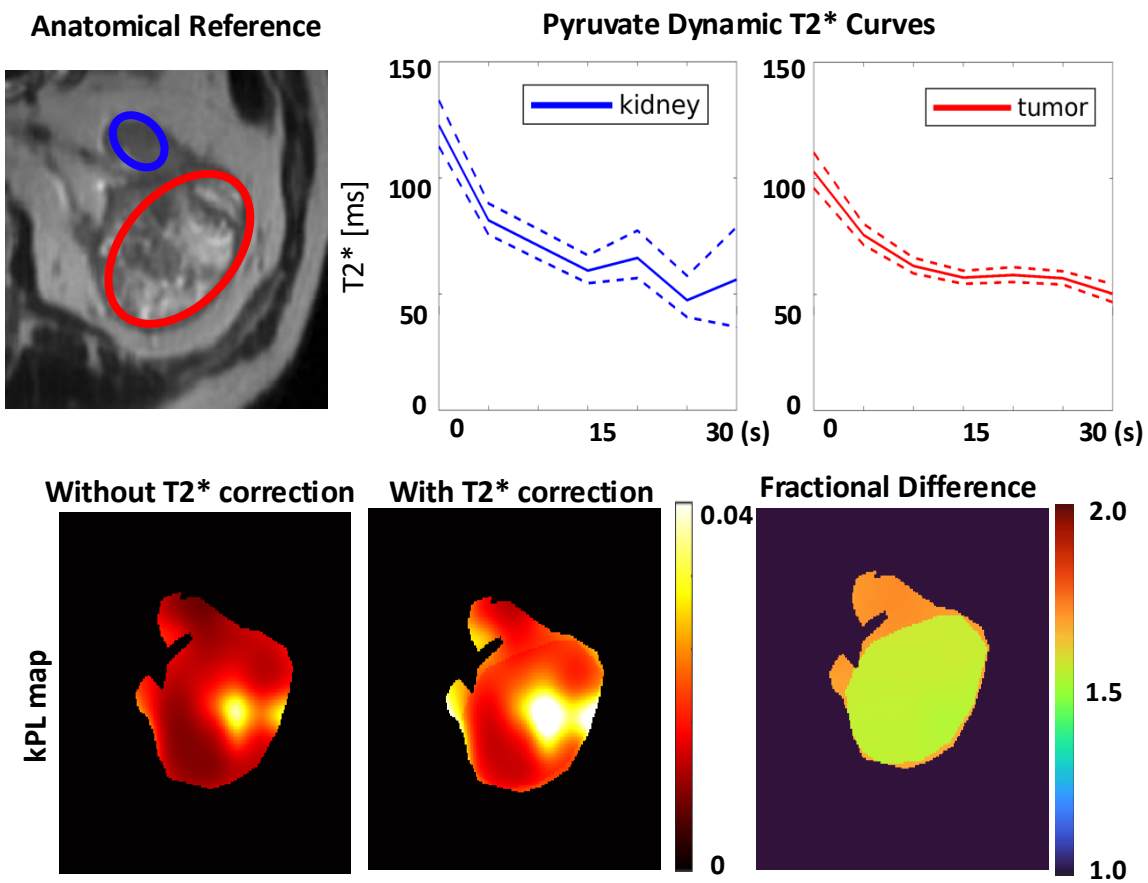
# C13 sequences — Modular 2D GRE spiral sequence

## Application for metabolite-specific multi-echo GRE seq —— optimizing quantification

### Brain Study Quantitative Maps



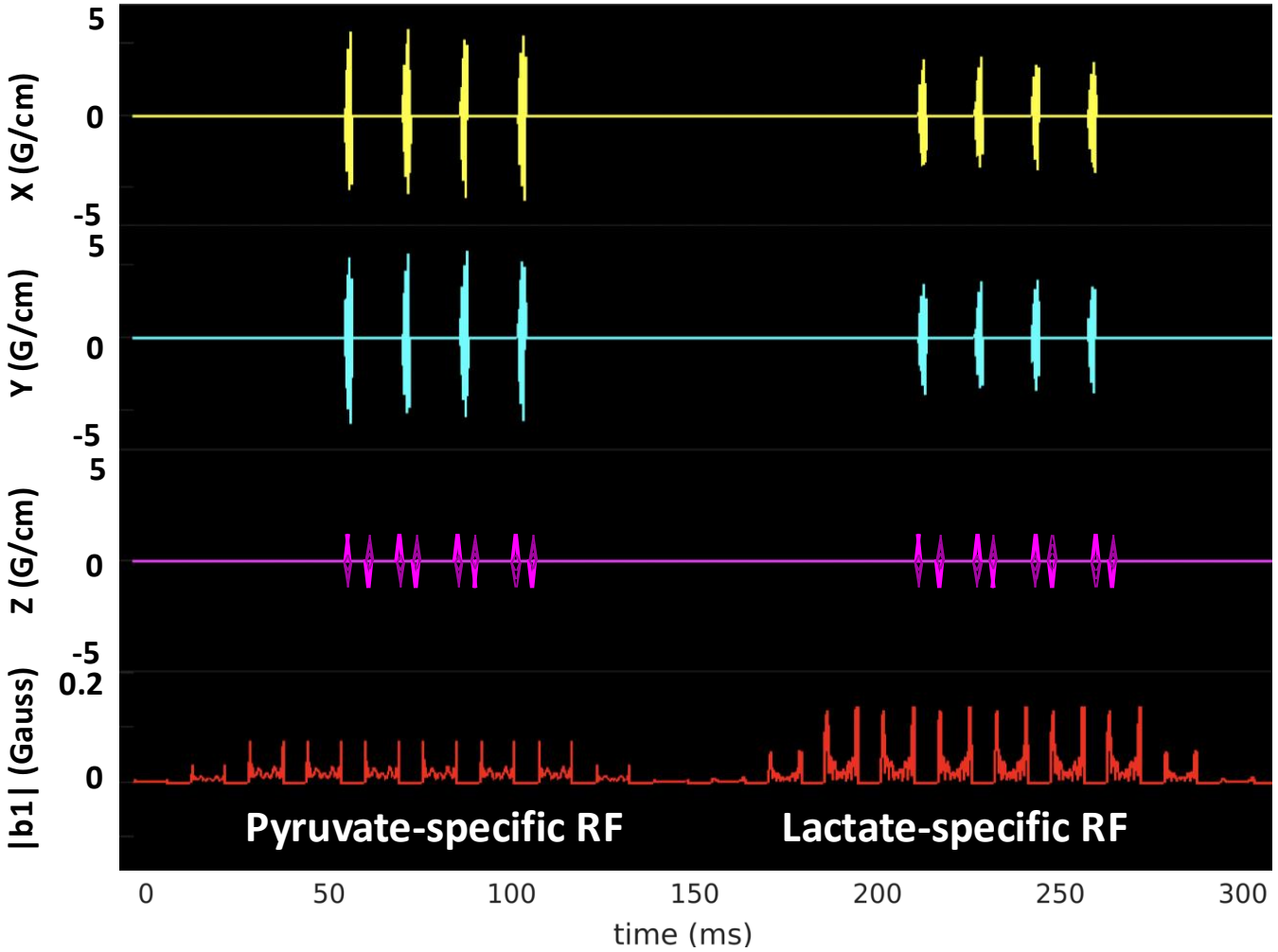

### Renal Cell Carcinoma Quantitative Maps




https://doi.org/10.1002/mrm.29942

# C13 sequences — Modular 3D MS-bSSFP sequence

## Different RF waveforms and TRs for different metabolites (different relative freq. shift)



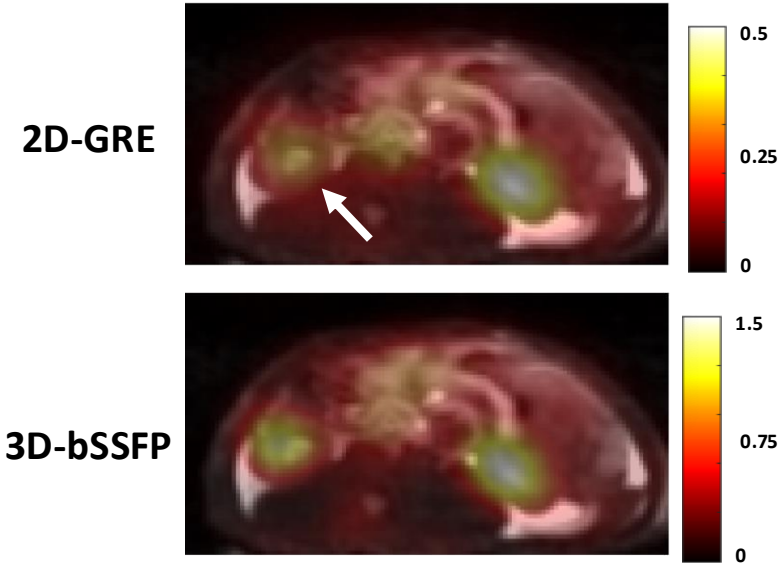

### Set sequence parameters

```
para = [];
para.nmet = 2; % number of metabolites
para.met_name = ['pyr'; 'lac']; % order of metabolites
para.freqshift = [0, 395]; % Hz, frequency shift
para.res = [10, 15]*1e-3; % m, resolution
para.nx = 45;  % matrix size
para.sth = 20e-3;  % m, slice thickness
para.nz = 16;  % number of slice
para.nT = 1; % number of time points
para.FA = [40, 60]; % degree, flip angle for each met
para.interleave = 5;
para.cat_FA = [0.05, 0.12, 0.38, 0.69, 0.93, 1];
para.tem_res = 5; % s, temporal resolution
```

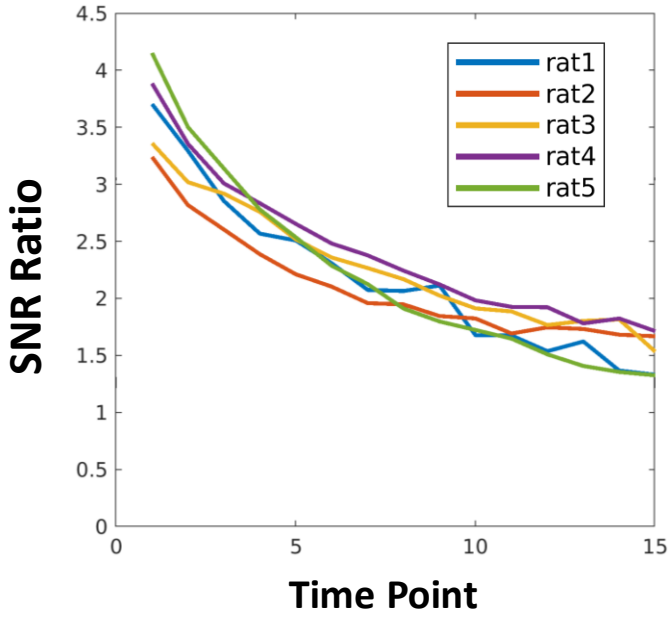
# C13 sequences — Modular 3D MS-bSSFP sequence

**3D MS-bSSFP provides higher SNR, but requires additional Series for different metabolites.**

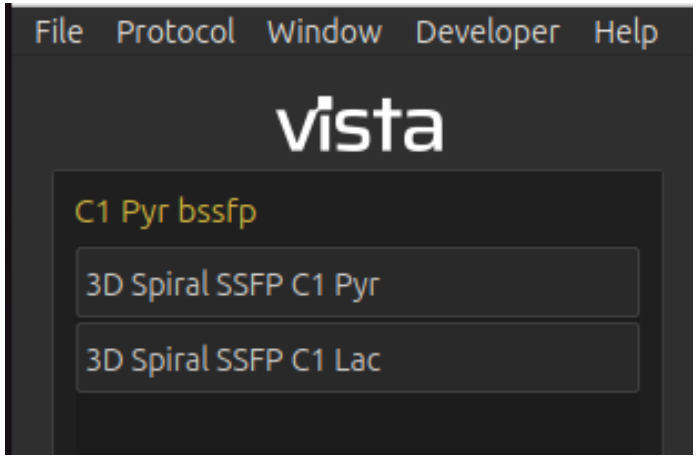
**Lactate in rat kidney**

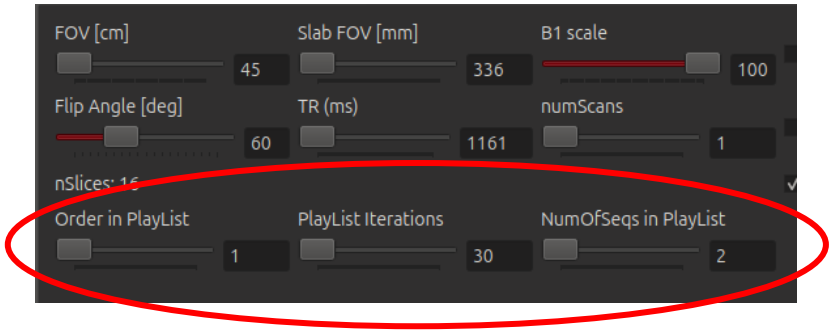



**3D-bSSFP to 2D-GRE**




**RTHawk Protocol**

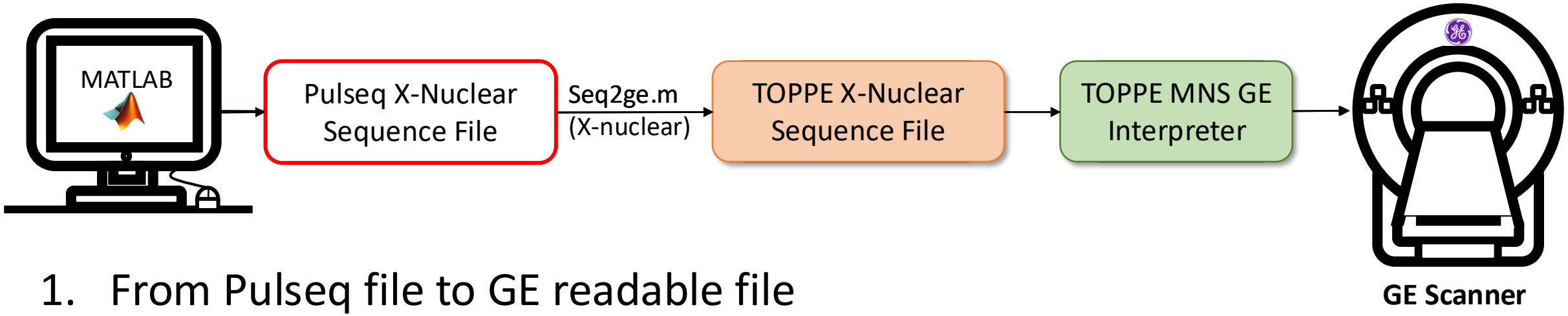



**~0.15s for switching series**

https://doi.org/10.1016/j.jmr.2023.107518

# TOPPE MNS




1. From Pulseq file to GE readable file

   Frequency offset looping; X-nuclei related scaling…

2. From sequence file to scanner run

   Adding GE built-in functions (broadband), exciter/receiver modification, RF/B1 scaling, nucleus choice on the user interface panel…

# TOPPE MNS

## Instruction of TOPPE MNS GE Interpreter

1. 3D Localizers
2. Frequency and transmit gain (TG) calibration by **fidall sequence**
3. Run Pulseq sequence with TOPPE MNS psd

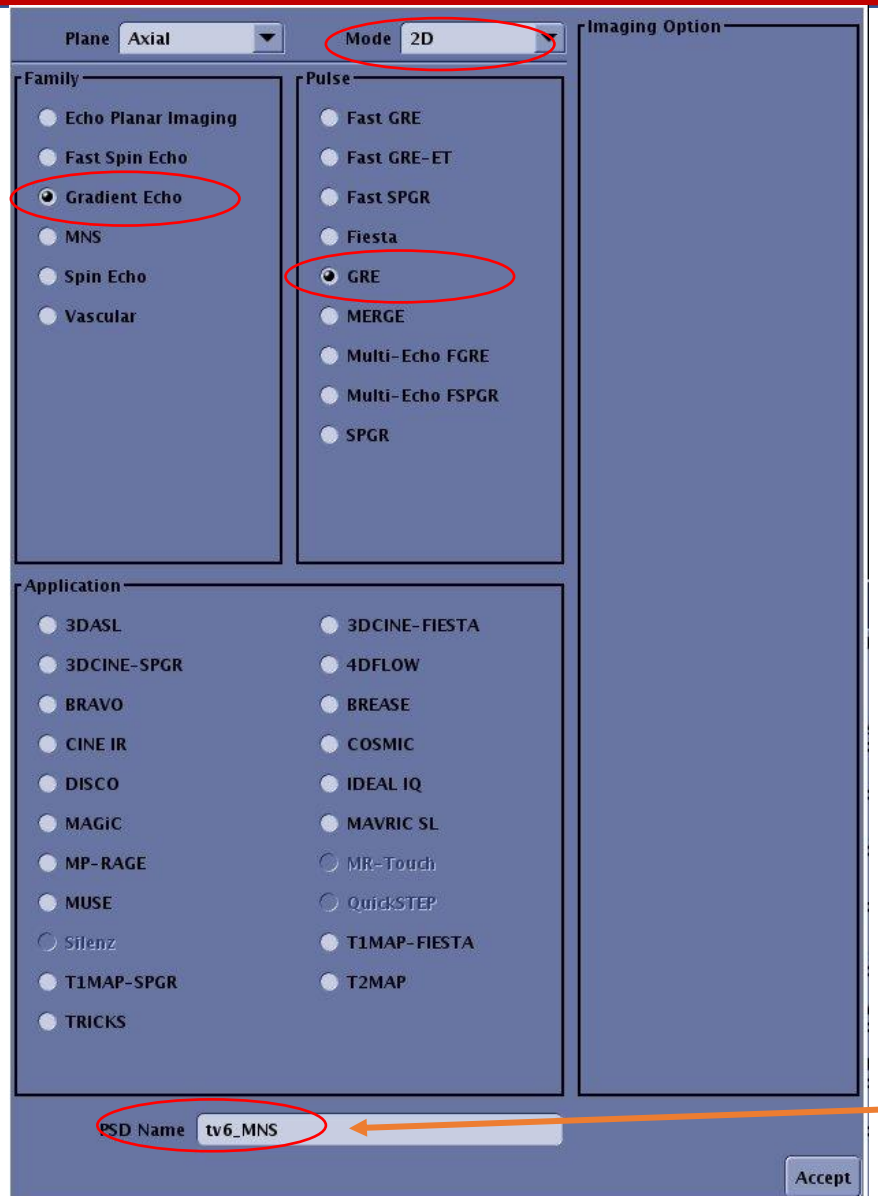
   Duplicate and setup the 'Localizer' sequence, add TOPPE MNS psd (same as the steps of TOPPE psd)

**fidall Sequence** – designed by **Rolf Schulte** (rolf.schulte@gehealthcare.com)
Version DV26.0: 1) spectrum prescan
Version MR30.0 or higher: 1) spectrum prescan, 2) MNS prescan, 3) Bloch-Siegert B1 mapping

# TOPPE MNS — Sequence Setting

User Interface:

Imaging options




For ALL sequences, select 2D-Gradient Echo-GRE. These settings are always the same, independent of the actual Pulseq scan you wish to run

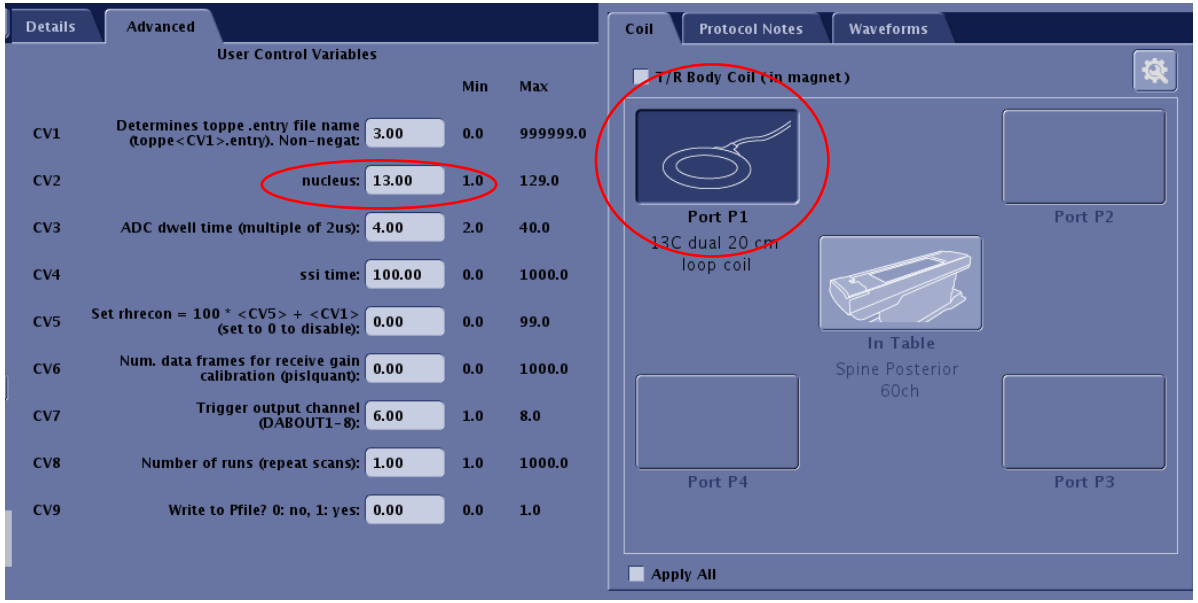
For pge2, just typing ‘pge2’ here

# TOPPE MNS — Sequence Setting

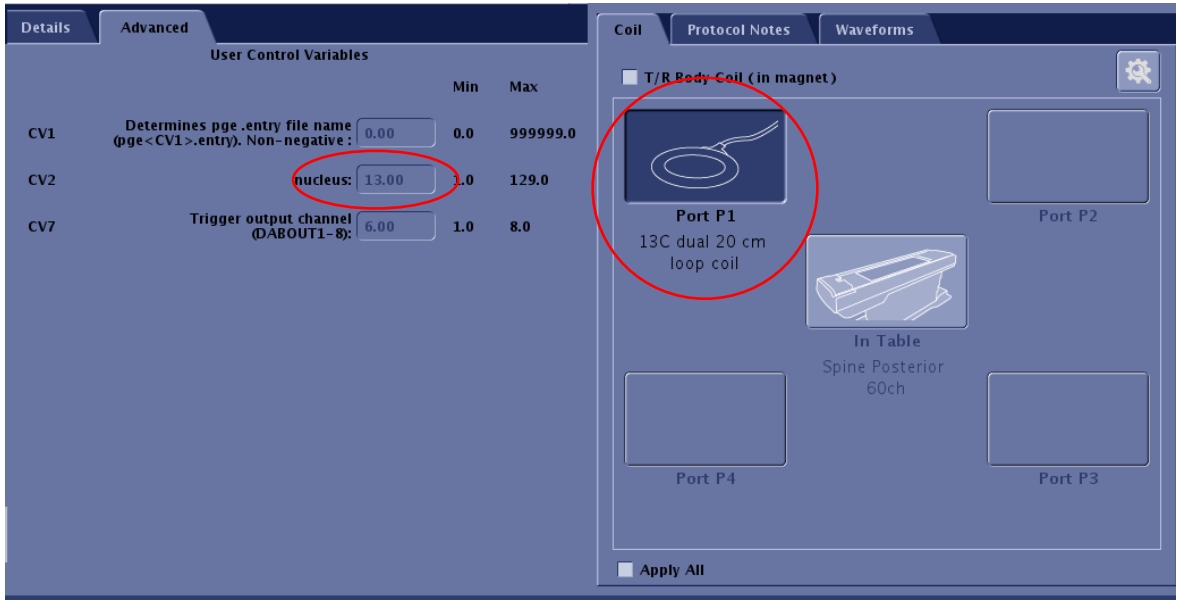
## User Interface —— Advanced Tab:

**Set number of nuclei and choose X-nuclear coil (e.g.: 13 and carbon coil)**

**tv6**

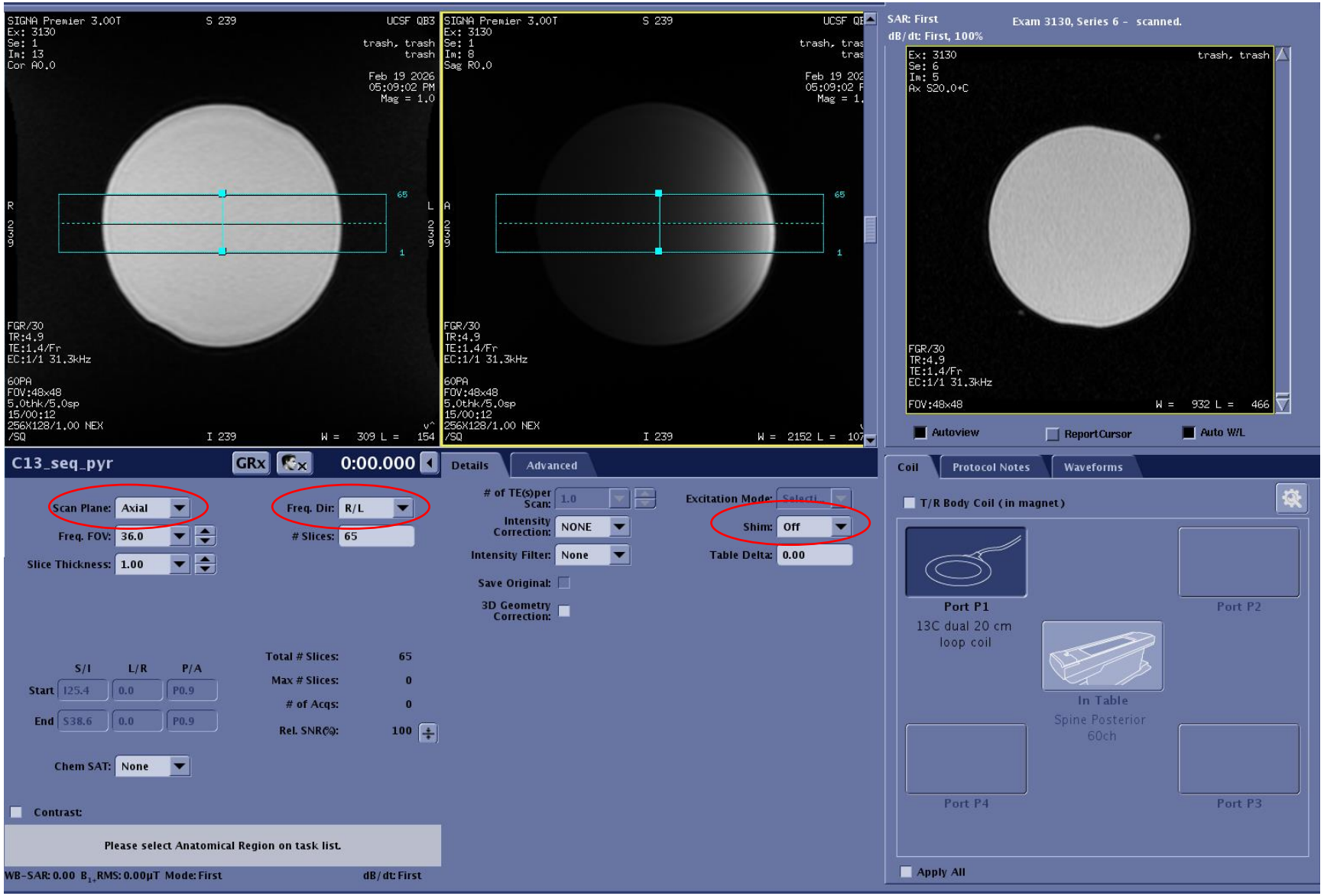



**pge2**



# TOPPE MNS — Sequence Setting

User Interface:



# TOPPE MNS — B1 Scaling

$$Flip\ angle:\ \alpha = \gamma \int B_1(t)dt$$

$$Gyromagnetic\ ratio:\ {}^{\gamma_{13_C}}/_{\gamma_{1_H}} \approx 1/4$$

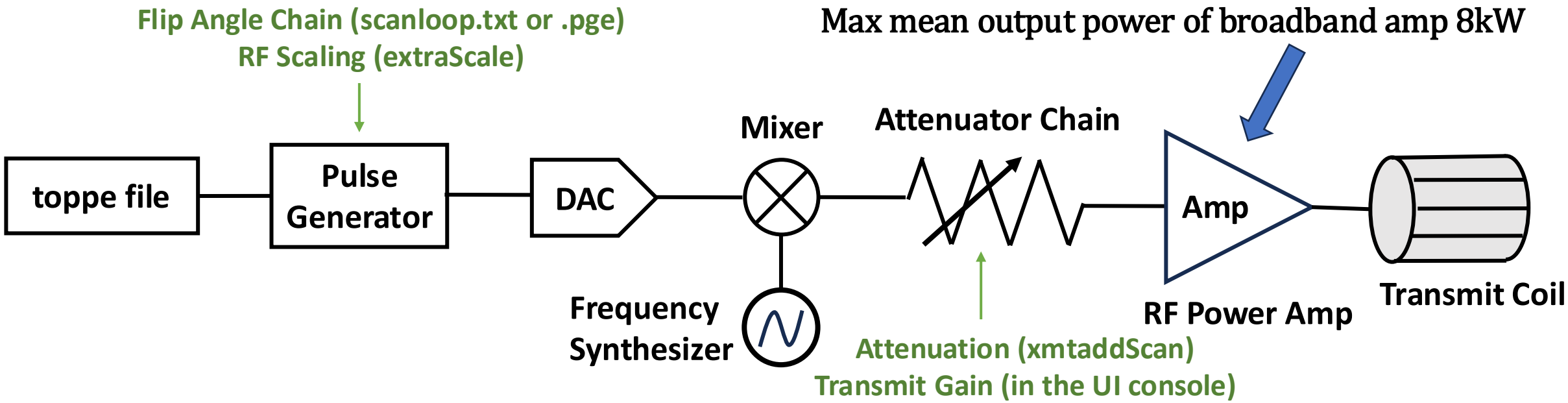
→ X-nuclear coils always support higher $B_1$

↓

$$P \propto B_1^2$$

↓

Max mean output power of broadband amp 8kW



# TOPPE MNS — B1 Scaling

$$TG_{toppe} = TG_{fidall} + 200 * \log_{10}\left(\frac{maxB1_{toppe}}{maxB1_{fidall}}\right)$$

**tv6:**

```
sysGE = toppe.systemspecs('maxGrad', 5, ...    % G/cm
    'maxSlew', 20, ...                         % G/cm/ms
    'maxRF', 0.15, ...                         % Gauss. Must be >= peak RF in sequence.
    'maxView', 120, ...                        % Determines slice/view index in data file
    'rfDeadTime', 100, ...                     % us
    'rfRingdownTime', 60, ...                  % us
    'adcDeadTime', 40, ...                     % us
    'psd_rf_wait', 148, ...                    % RF/gradient delay (us)
    'psd_grd_wait', 156);                      % ADC/gradient delay (us)

seq2ge('gre3d.seq', sysGE, 'gre3d.tar');
```

This value should be lower than maxB1 of coil (set in coil config file)

Error: ‘minseqrfamp failed’

**pge2:**

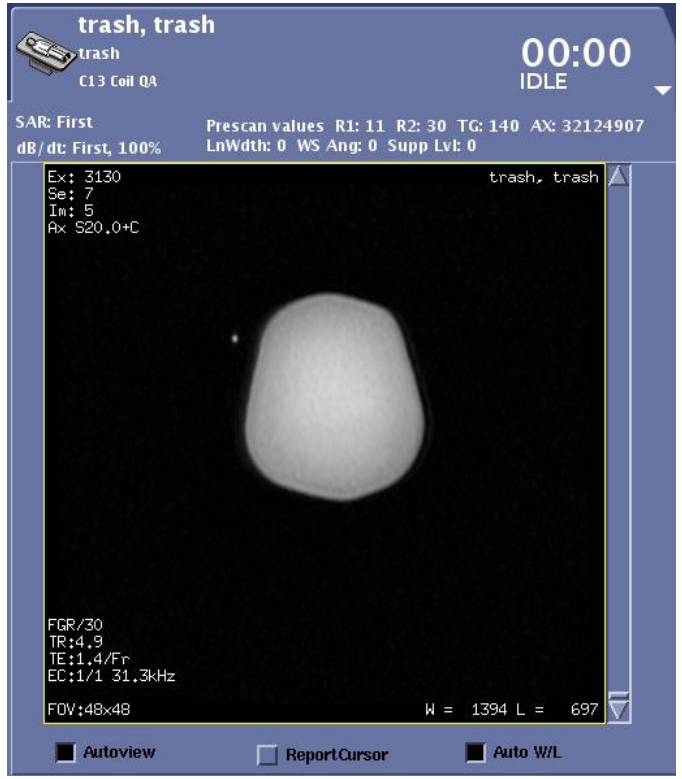
**X-nuclear sequences: maxB1 = maxB1 in the sequence**
**Proton sequences: maxB1 in sequence < 0.1474 Gauss, maxB1 = 0.1474 Gauss**
**maxB1 in sequence > 0.1474 Gauss, maxB1 = maxB1 in sequence**

# TOPPE MNS — B1 Scaling

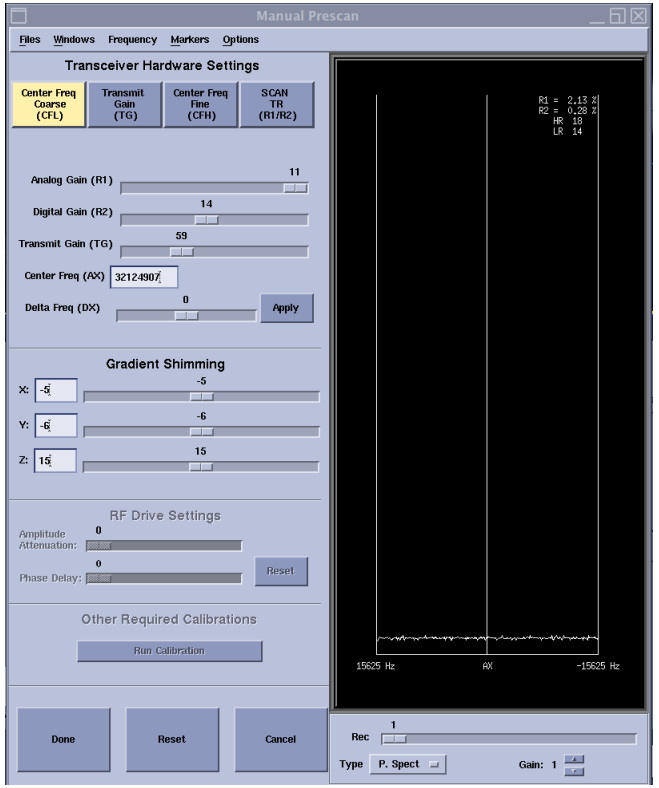
$$TG_{toppe} = TG_{fidall} + 200 * \log_{10}\left(\frac{maxB1_{toppe}}{maxB1_{fidall}}\right)$$

**Dual Tuned $^{1}H/^{13}C$ Head Coil (Rapid Biomedical)**

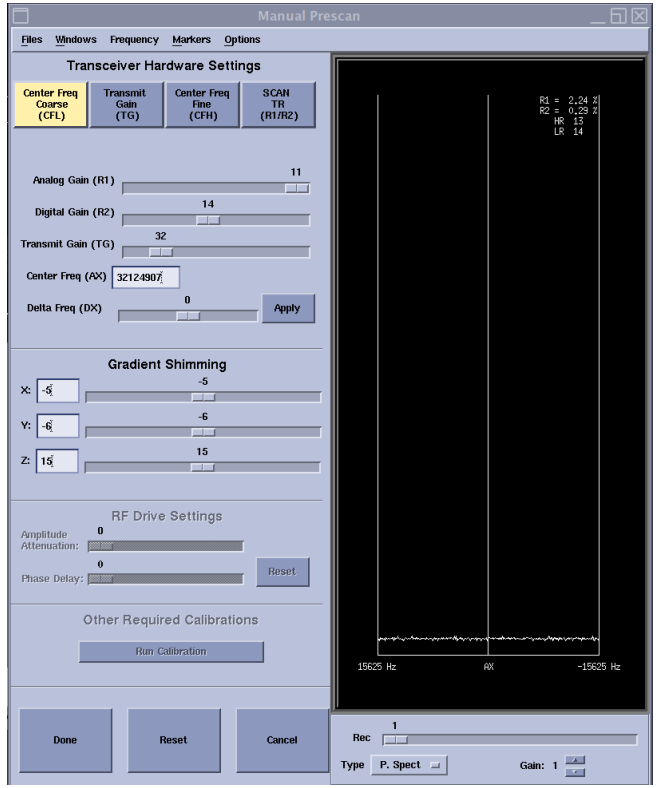
$TG_{fidall} = 140, B1 = 0.51\ Gauss$




$TG_{tv6} = 59, B1 = 0.2\ Gauss$



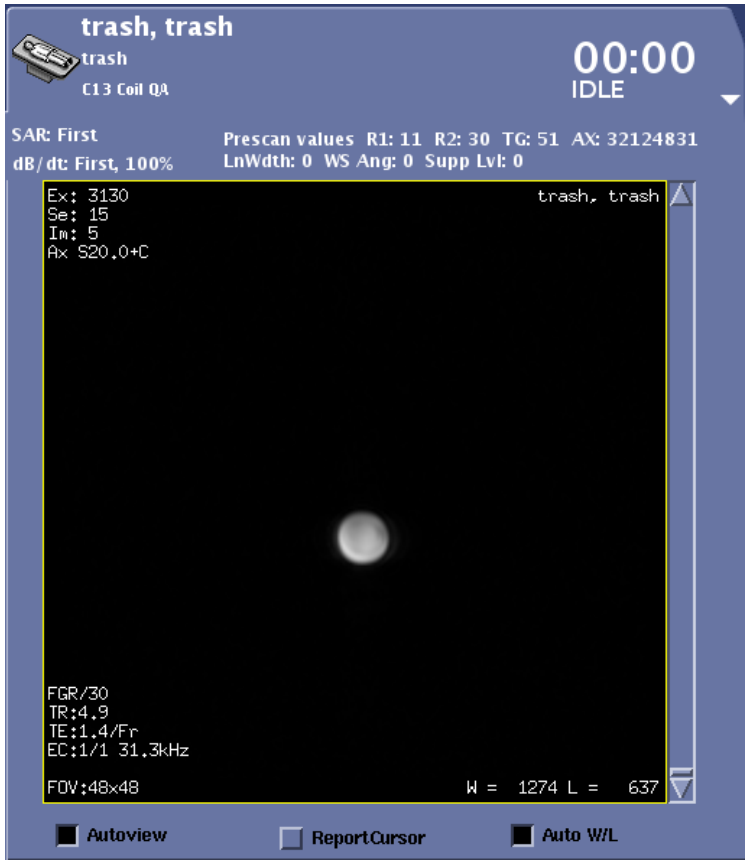

$TG_{pge2} = 32, B1 = 0.147\ Gauss$



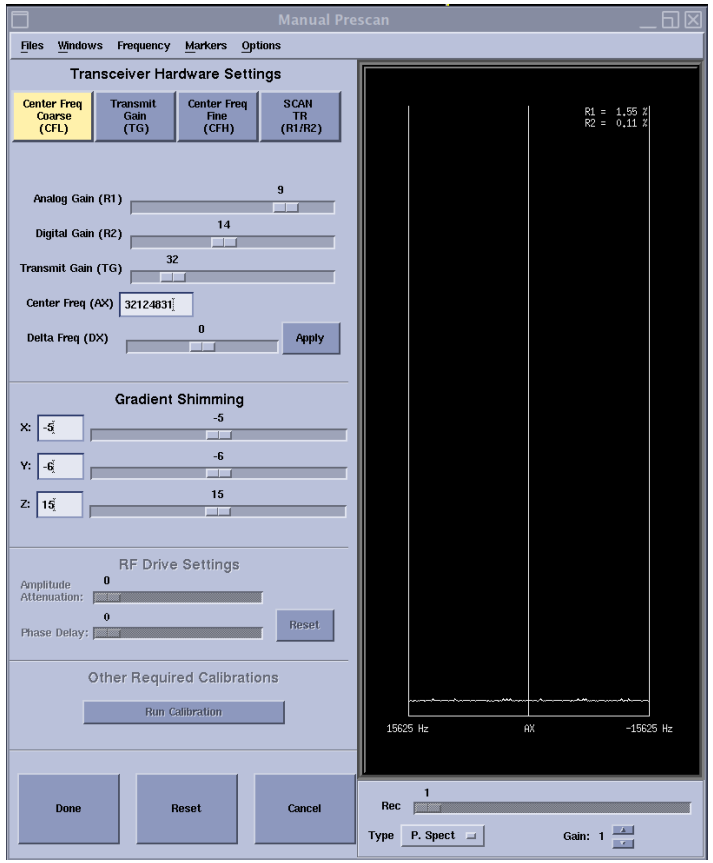
# TOPPE MNS — B1 Scaling

$$TG_{toppe} = TG_{fidall} + 200 * \log_{10}\left(\frac{maxB1_{toppe}}{maxB1_{fidall}}\right)$$

$TG_{fidall} = 51, B1 = 0.25\ Gauss$




$TG_{tv6} = 32, B1 = 0.2\ Gauss$




Dual-tuned $^{1}H/^{13}C$ transceiver birdcage coil

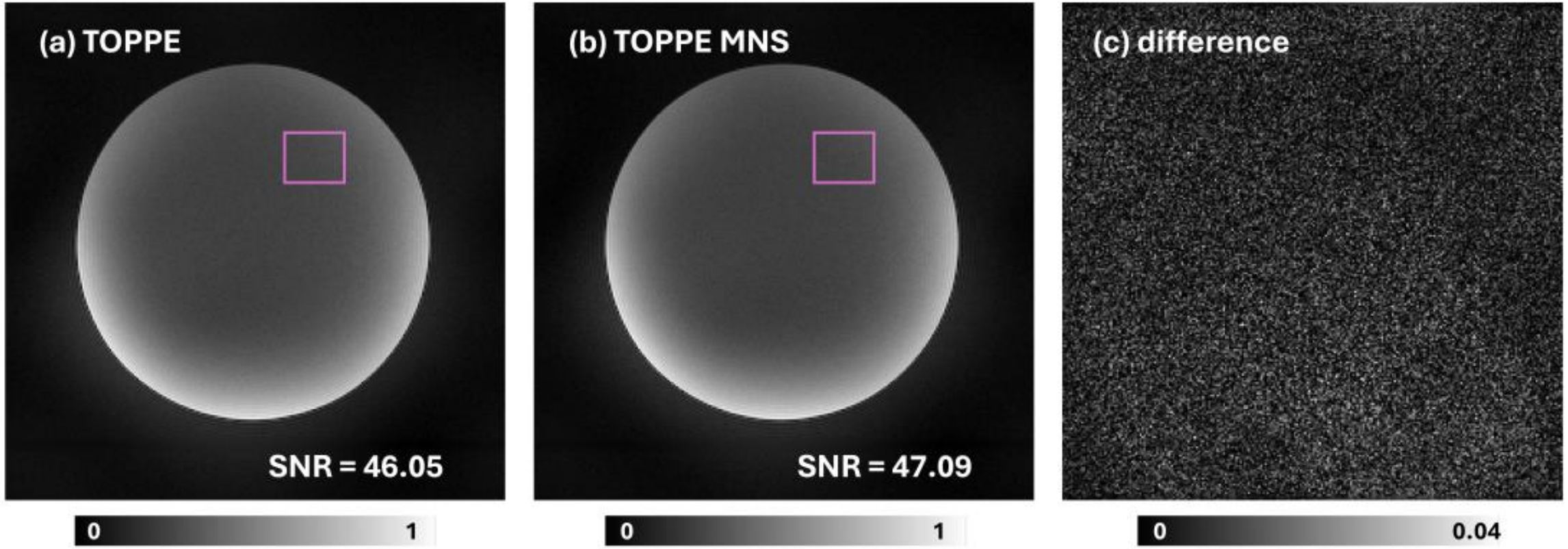
R1/R2

Version < MR30.0, keep same

Version >= MR30.0, R2 <= 15

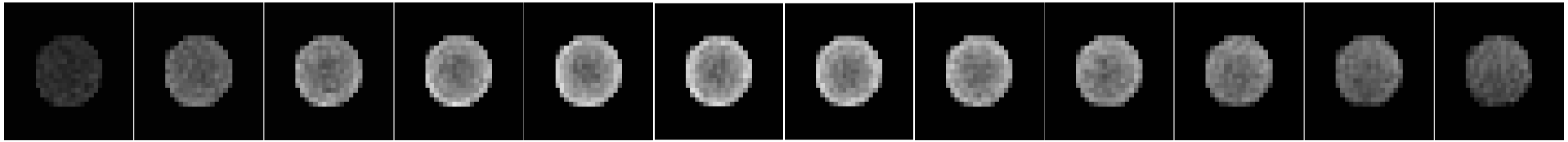
# Results – Proton Functionality Evaluation

**2D GRE sequence executed by TOPPE and TOPPE MNS GE interpreters**



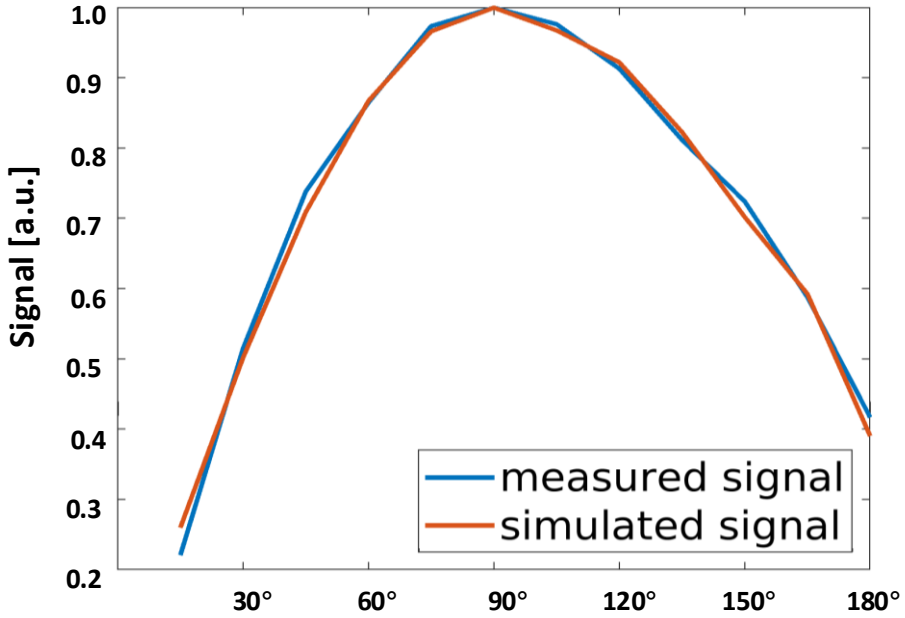
# Results – RF Power Evaluation

**(a) Images at different flip angle acquired by $^{13}C$ coil**

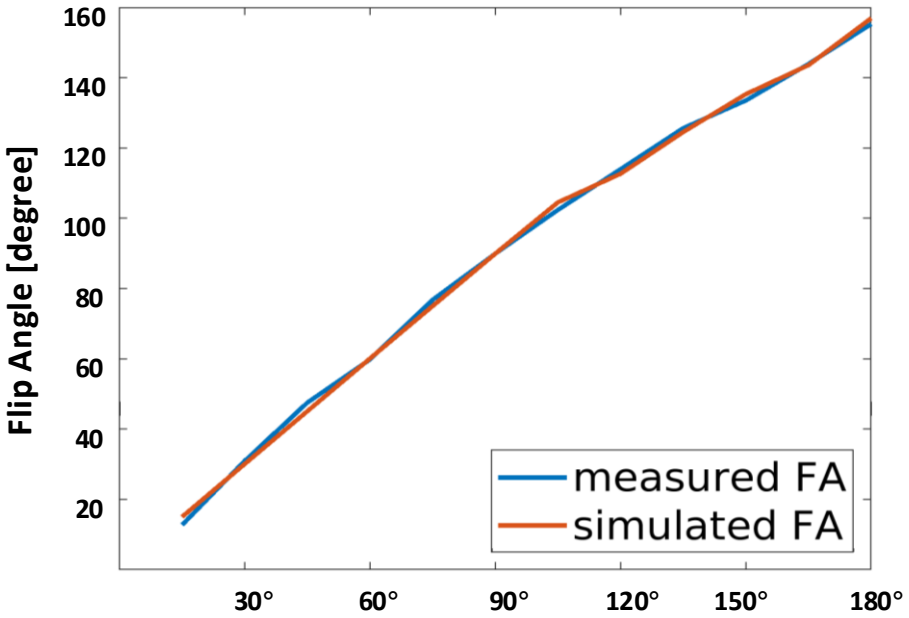


15° 30° 45° 60° 75° 90° 105° 120° 135° 150° 165° 180°

**(b) Signal curve**




**(c) Flip angles**



# Results – RF Power Evaluation

**Jon has shared a B1 calibration sequence (version: pge2) in the GitHub**
https://github.com/HarmonizedMRI/SequenceExamples-GE/tree/develop/pge2/b1CalSimple

## main.m

```
% Check the ceq object:
% Define hardware parameters, and
% check if 'ceq' is compatible with the parameters in 'sys'
psd_rf_wait = 58e-6;  % RF-gradient delay, scanner specific (s)
psd_grd_wait = 60e-6; % ADC-gradient delay, scanner specific (s)
b1_max = 0.25;         % Gauss
g_max = 5;             % Gauss/cm
slew_max = 20;         % Gauss/cm/ms
gamma = 4.2576e3;      % Hz/Gauss
sys = pge2.getsys(psd_rf_wait, psd_grd_wait, b1_max, g_max, slew_max, gamma);
pge2.validate(ceq, sys);
```

Updating the gamma

## writeflip.m

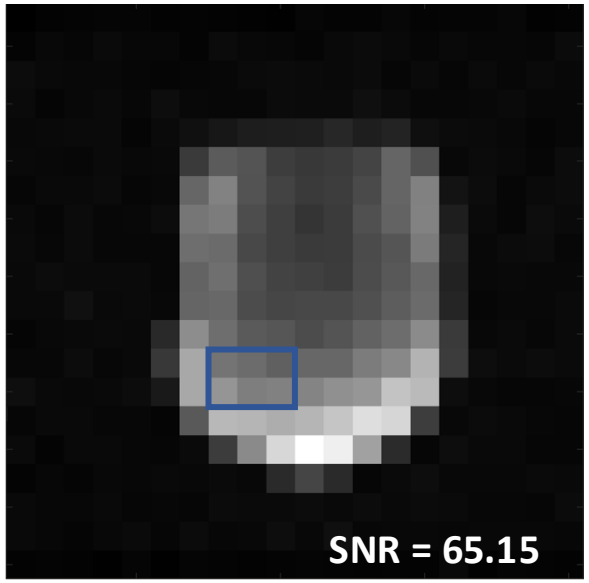
Increasing pulse duration!!! (maxB1 no more than 0.25 Gauss)

$$e.g.: duration * {}^{\gamma_{1_H}}/_{\gamma_{X-nuclei}}$$

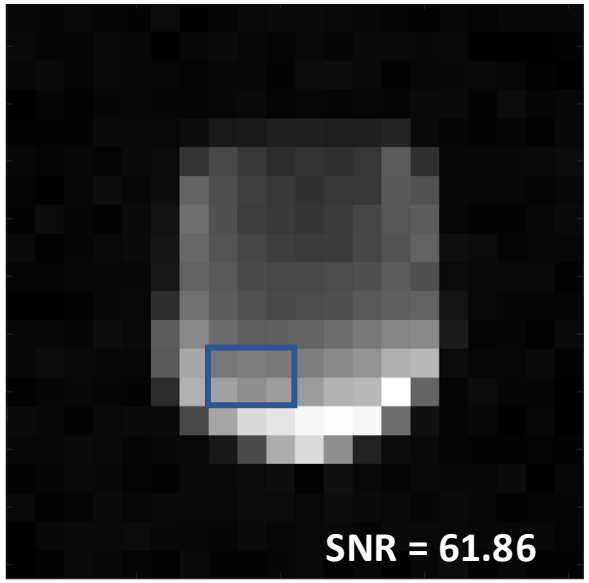
```
% Create 180-degree slice selection pulse and gradient
[rf, gz] = mr.makeSincPulse(pi, 'Duration', 4e-3, ...
    'SliceThickness', sliceThickness, 'apodization', 0.42, ...
    'use', 'excitation', ...
    'timeBwProduct', 4, 'system', sys);
gzReph = mr.makeTrapezoid('z', 'Area', -gz.area/2, 'Duration', 1e-3, 'system', sys);
```

# Results – Quality Control

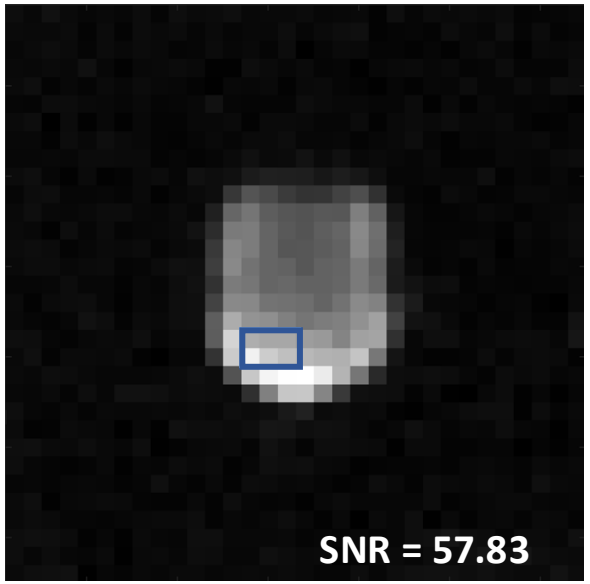
**Symmetric EPI**



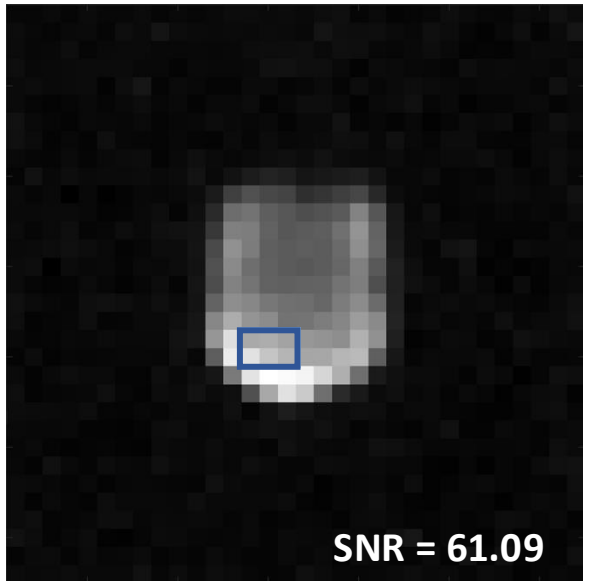

**Pulseq EPI**




**fidall Spiral**




**Pulseq Spiral**



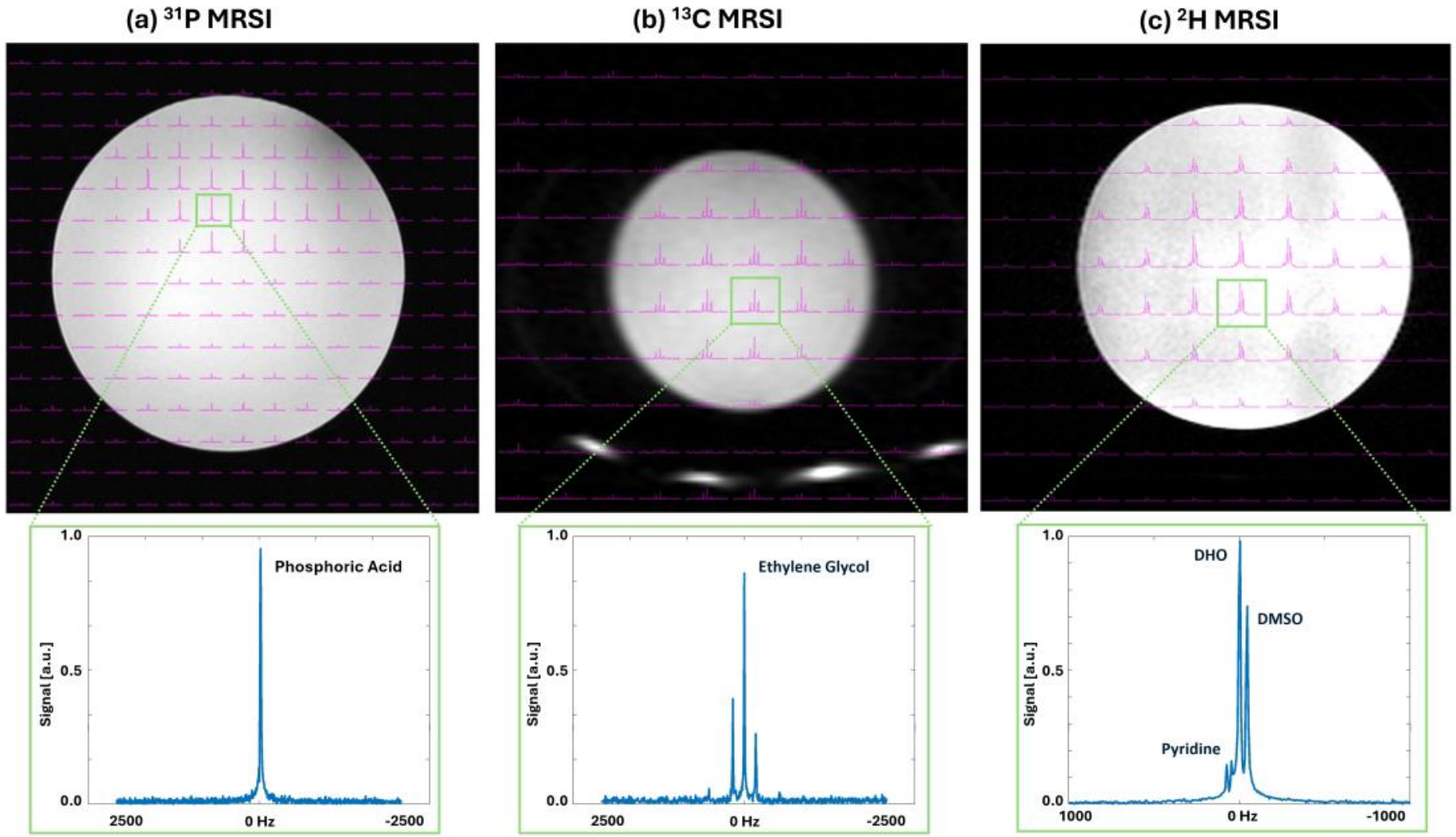

**Acquisition parameters of EPI**

resolution = 1.5x1.5 $cm^2$
slice thickness = 2 cm
FOV = 24x24 $cm^2$
TE/TR = 25.6/1000 ms
flip angle = 60°
NEX = 120
echo spacing (ESP) = 1.3 ms , BW = 8.93 kHz
R1/R2: 9/14

**Acquisition parameters of Spiral**

resolution = 1.5x1.5 $cm^2$
slice thickness = 2 cm
FOV = 48x48 $cm^2$
TE/TR = 12.6/1000 ms
flip angle = 60°
NEX = 120
R1/R2: 10/30 (fidall with EDR), 9/14 (tv6)
EDR: extended dynamic range

# Results – X-Nuclear Acquisition Validation



# Results – Multi-Version Study

**$^{13}$C-Urea Needle Phantom**

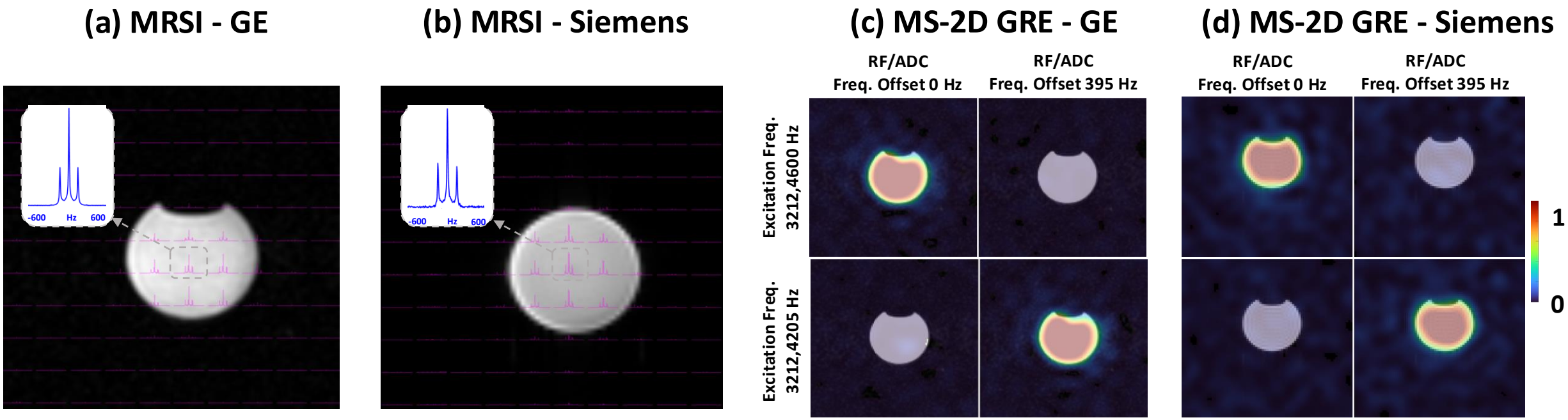
Version 1,UCSF: Premier, GE Healthcare, MR30.1_R01

Version 2, University of Maryland: MR750w, GE Healthcare, DV26.0_R02

TE1 = 12.6 ms TE2 = 26.9 ms TE3 = 41.2 ms TE4 = 55.5 ms TE5 = 69.8 ms TE6 = 84.1 ms

Same sequence files were generated by the modular GRE spiral seq. script, and then run on the GE scanner with version MR30.1_R01 (@UCSF) and GE scanner with version DV26.0_R02 (@UMD, **Dirk Mayer**).

# Results – Multi-Vendor Study



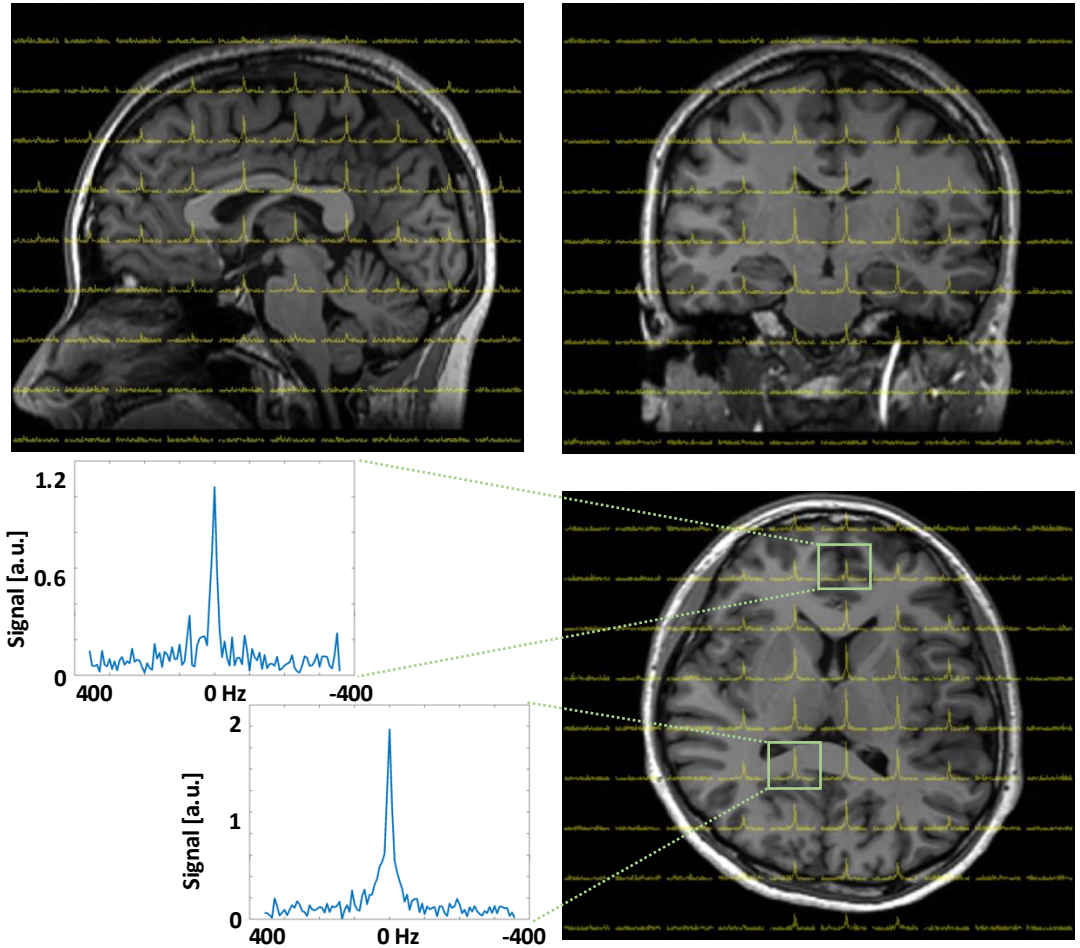

Sequence files were generated by the same modular GRE spiral seq. script, and then executed on the GE scanner (@UCSF) and Siemens scanner (@DKFZ, **Andreas Korzowski**).

GE 3T Premier MR, version MR30.1_R01
Siemens 3T clinical PET/MR system, version VE11 IDEA version

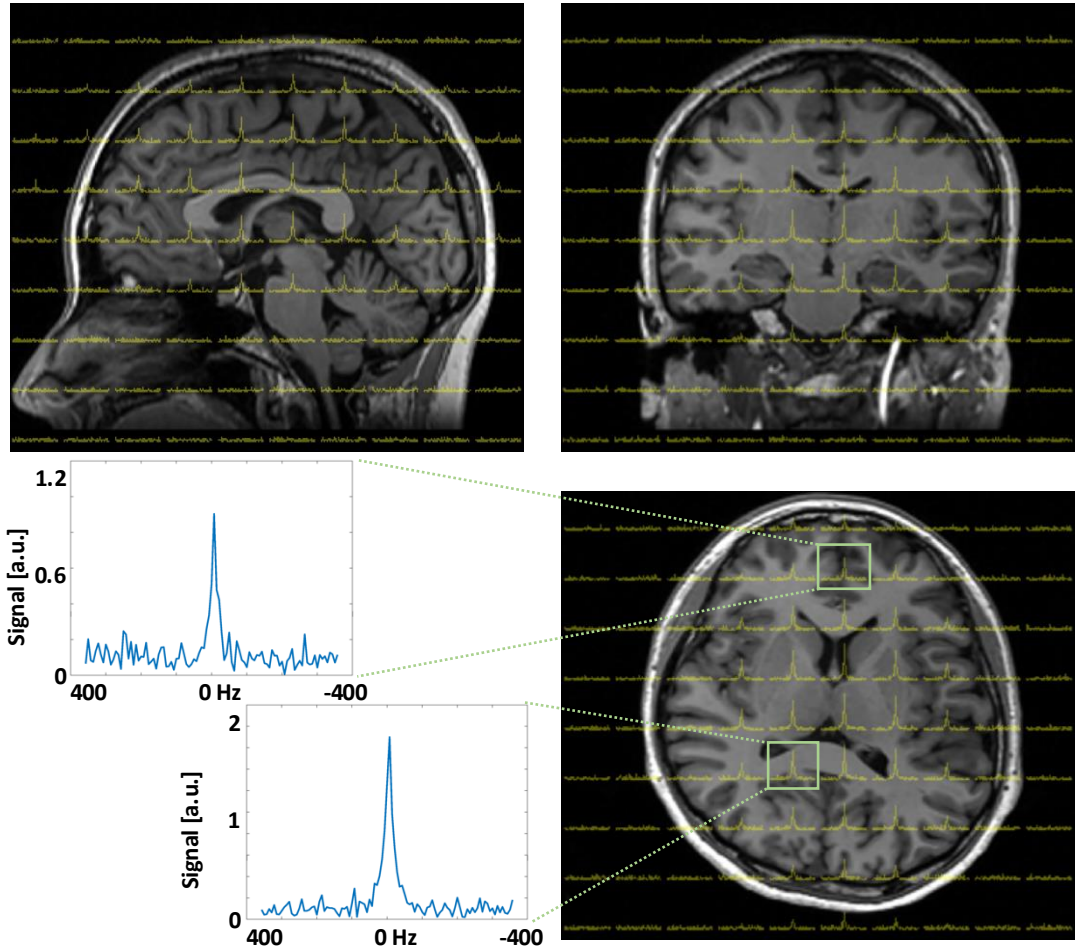
# Results – Higher Order Shimming Compatibility

## 3D MRSI $^{2}$H spiral sequence [1]

(a) With Second Order Shimming

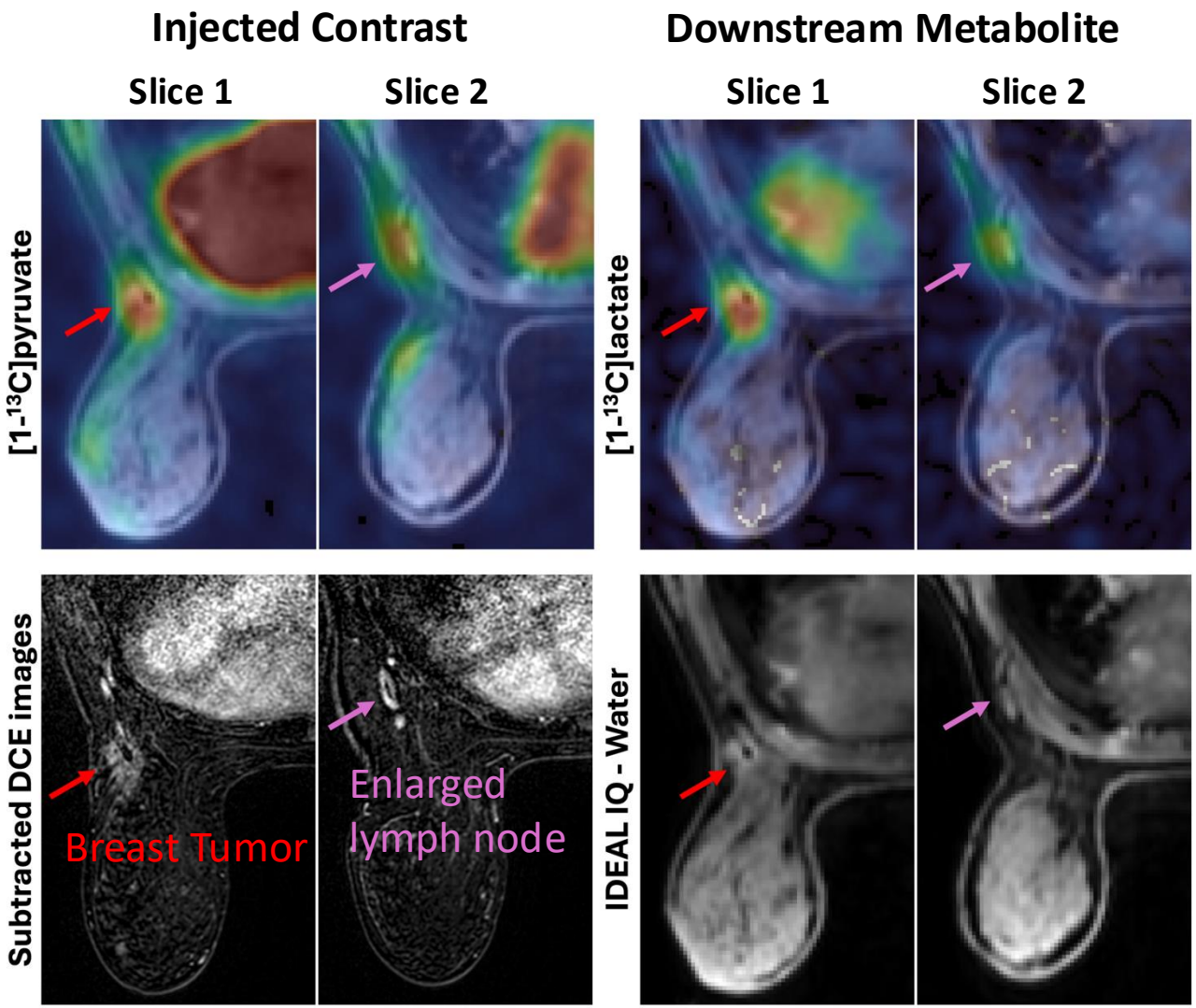



(b) With Linear Shimming




[1] https://doi.org/10.1186/s41747-024-00426-4

# Results – Hyperpolarized $^{13}C$ Human BCa Study



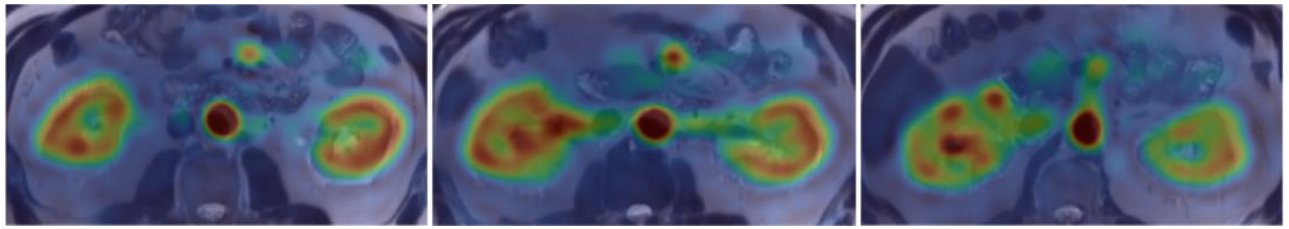

**Acquisition Parameters**

Sequence: Modular 2D GRE spiral sequence
PSD: tv6_MNS
Resolution: 1x1 $cm^2$
Slice thickness: 2.1 cm
Flip angle: 20° (pyruvate), 30° (lactate)
Transmit Gain: 99 (maxB1: 0.2 Gauss)
Receive Gain (R1/R2): 9/14

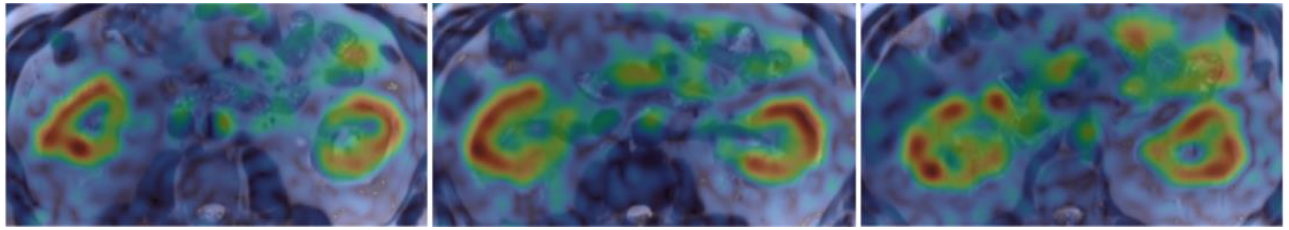
**4-Rung Transmit Coil (RAPID Biomedical)**

# Results – Hyperpolarized $^{13}C$ Human RCC Study

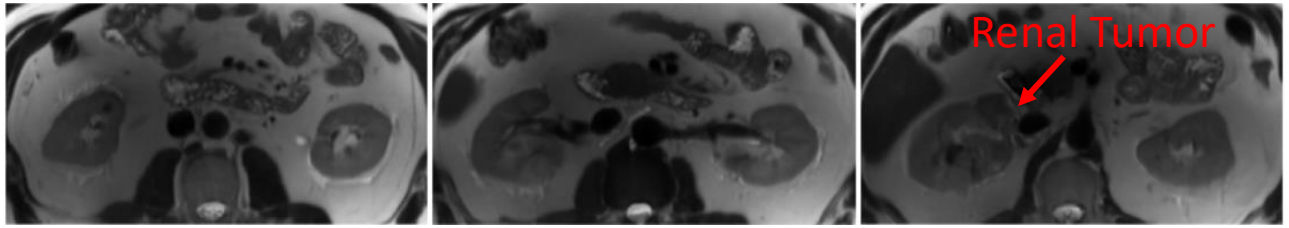
**Injected contrast (Pyruvate)**



**Downstream met. (Lactate)**



**T2w-FSE (anatomical reference)**




**Slice 1** **Slice 2** **Slice 3**

**Acquisition Parameters**

Sequence: Modular 3D MS-bSSFP sequence
PSD: pge2
Resolution: 1x1 $cm^2$
Slice thickness: 2 cm
Flip angle: 40° (pyruvate), 60° (lactate)
Transmit Gain: 82 (maxB1: 0.15 Gauss)
Receive Gain (R1/R2): 9/14

**Semi-flexible Quadrature Transmit and Receive (QTAR) Array (Clinical MR Solutions)**

# Acknowledgement:

We really thank Dr. **Jon-Fredrik Nielsen** in University of Michigan for building the TOPPE interpreter, letting us have the chance to run X-nuclear Pulseq sequence on GE scanner.

TOPPE interpreter with MNS (Pulseq on GE v1, tv6):
https://github.com/jfnielsen/TOPPEpsdSourceCode/tree/develop

MNS function has been combined into latest version (Pulseq on GE v2, pge2):
https://github.com/jfnielsen/TOPPEpsdSourceCode/tree/tv7/v7

# Acknowledgement:

**Feel free contact us for any questions on the MNS functions in TOPPE.**

**Di Cui**
**Di.Cui@ucsf.edu**

**Xiaoxi Liu**
**Xiaoxi.Liu@ucsf.edu**

Pulseq C13 sequences: https://github.com/Xiaoxi-Liu/C13_sequences_Pulseq
Paper: https://doi.org/10.1002/mrm.30509

# Acknowledgement —— Our special collaborators

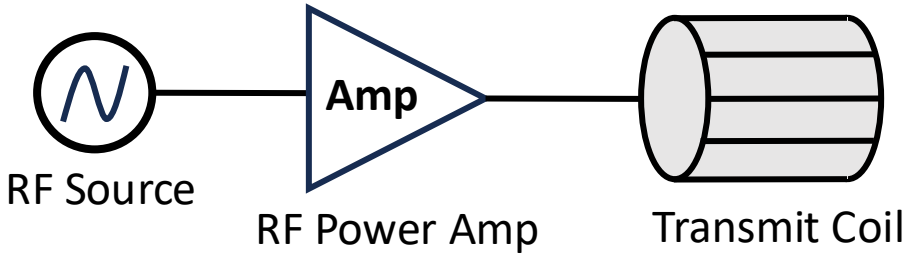
**GE 3T MRI scanner —— born Feb 2024 @ UCSF**

- GE SIGNA™ Premier 3T, version: MR30.1
- Dual-tuned $^{1}H/^{13}C$ transceiver birdcage coil with high limiter to limit the power from RFPA (Designed and built by **Lucas Carvajal**)

VISTA RTHawk → $^{13}C$ seq → Pulseq **+ TOPPE**

@GE 3T MR750

@GE 3T Premier



# Acknowledgement:


## UCSF

Xiaoxi Liu
Di Cui
Peder Larson
Changhua Mu
Lucas Carvajal
Duan Xu
Jeremy Gordon
Daniel Vigneron
Robert Flavell
Zhen Wang
Bonnie Joe

## University of Maryland

Dirk Mayer

## DKFZ

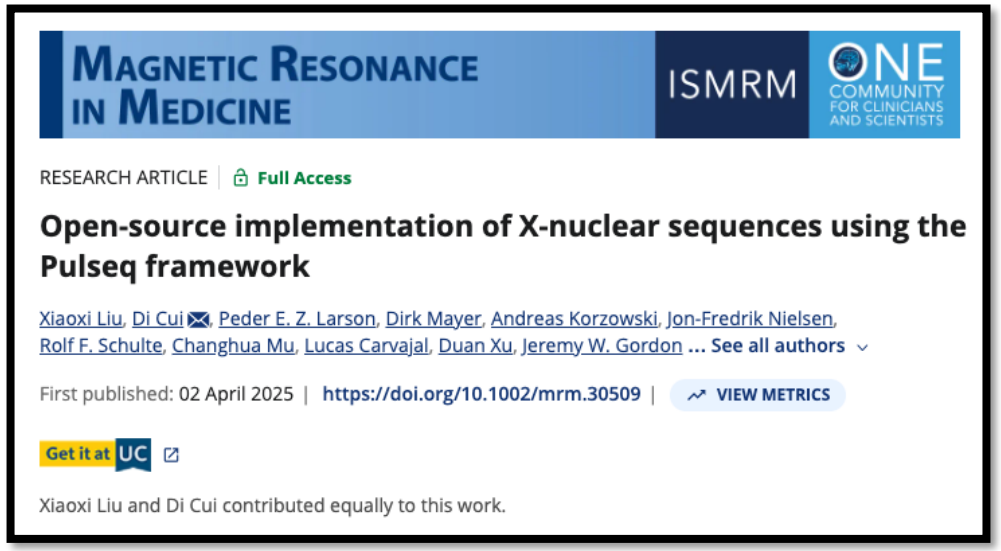
Andreas Korzowski

## GE HealthCare

Rolf Schulte

## University of Michigan

Jon-Fredrik Nielsen

MAGNETIC RESONANCE IN MEDICINE | ISMRM | ONE COMMUNITY FOR CLINICIANS AND SCIENTISTS

RESEARCH ARTICLE | Full Access

**Open-source implementation of X-nuclear sequences using the Pulseq framework**

Xiaoxi Liu, Di Cui, Peder E. Z. Larson, Dirk Mayer, Andreas Korzowski, Jon-Fredrik Nielsen, Rolf F. Schulte, Changhua Mu, Lucas Carvajal, Duan Xu, Jeremy W. Gordon ... See all authors

First published: 02 April 2025 | https://doi.org/10.1002/mrm.30509 | VIEW METRICS

Get it at UC

Xiaoxi Liu and Di Cui contributed equally to this work.

**Enjoy your first X-nuclear Pulseq sequence!**

# Q&A

**Thank you!**